Test Series No.

01

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

ANSWER KEY

1. (1)	46. (1)	91. (3)	136. (1)
2. (2)	47. (4)	92. (2)	137. (2)
3. (1)	48. (1)	93. (3)	138. (3)
4. (1)	49. (1)	94. (1)	139. (2)
5. (1)	50. (1)	95. (1)	140. (3)
6. (3)	51. (1)	96. (4)	141. (1)
7. (1)	52. (4)	97. (4)	142. (4)
8. (3)	53. (2)	98. (4)	143. (4)
9. (2)	54. (1)	99. (3)	144. (1)
10. (4)	55. (3)	100. (2)	145. (3)
11. (1)	56. (1)	101. (1)	146. (3)
12. (2)	57. (1)	102. (1)	147. (4)
13. (3)	58. (3)	103. (3)	148. (1)
14. (2)	59. (4)	104. (2)	149. (3)
15. (2)	60. (3)	105. (2)	150. (3)
16. (1)	61. (2)	106. (3)	151. (3)
17. (1)	62. (2)	107. (4)	152. (2)
18. (2)	63. (3)	108. (4)	153. (3)
19. (1)	64. (1)	109. (1)	154. (3)
20. (2)	65. (1)	110. (2)	155. (1)
21. (1)	66. (2)	111. (3)	156. (4)
22. (3)	67. (4)	112. (1)	157. (2)
23. (2)	68. (4)	113. (1)	158. (3)
24. (4)	69. (2)	114. (1)	159. (3)
25. (4)	70. (2)	115. (1)	160. (3)
26. (1)	71. (2)	116. (4)	161. (4)
27. (2)	72. (3)	117. (4)	162. (3)
28. (2)	73. (3)	118. (1)	163. (4)
29. (2)	74. (4)	119. (1)	164. (2)
30. (3)	75. (1)	120. (3)	165. (4)
31. (1)	76. (2)	121. (2)	166. (2)
32. (3)	77. (2)	122. (1)	167. (4)
33. (4)	78. (4)	123. (3)	168. (1)
34. (1)	79. (2)	124. (4)	169. (4)
35. (2)	80. (4)	125. (4)	170. (2)
36. (1)	81. (2)	126. (4)	171. (4)
37. (1)	82. (1)	127. (3)	172. (1)
38. (2)	83. (2)	128. (4)	173. (4)
39. (2)	84. (2)	129. (4)	174. (1)
40. (2)	85. (4)	130. (4)	175. (4)
41. (4)	86. (3)	131. (2)	176. (2)
42. (4)	87. (1)	132. (1)	177. (4)
43. (4)	88. (1)	133. (4)	178. (1)
44. (3)	89. (2)	134. (4)	179. (1)
45. (4)	90. (3)	135. (2)	180. (1)

1. (1)

$$y = x - \frac{x^2}{20} = x\left(1 - \frac{x}{20}\right)$$

On comparing with $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

$R = 20$ m

(NCERT Ed. 2025-26, 11th, Page No. 40)

2. (2)

The energy released (Q -value) is the difference between the total binding energy of the products and the total binding energy of the reactant.

Total initial Binding Energy (Reactant)

$= 240 \times 7.6$ MeV

Total final Binding Energy (Products)

$= 240 \times 8.5$ MeV

Energy Released (Q) = Final B.E. – Initial B.E.

$Q = 240 \times (8.5 - 7.6)$

$Q = 240 \times 0.9 = 216$ MeV

(NCERT Ed. 2025-26, 12th, Page No. 315)

3. (1)

In Bohr's model, Energy $E \propto m$ and radius $r \propto \frac{1}{m}$

If mass is doubled, $E_0 = 2 \times (-13.6) = -27.2$ eV

and $r_0 = \frac{a_0}{2}$.

(NCERT Ed. 2025-26, 12th, Page No. 299)

4. (1)

λ is wavelength in air

$$\frac{8\lambda D}{d} = \frac{17\lambda' D}{2d}$$

$$\Rightarrow \mu = \frac{\lambda}{\lambda'} = \frac{17}{16}$$

(NCERT Ed. 2025-26, 12th, Page No. 266)

5. (1)

Least Count (LC):

$$\text{Pitch} = \frac{\text{Distance moved}}{\text{No. of rotations}} = \frac{1 \text{ mm}}{2} = 0.5 \text{ mm}$$

$$LC = \frac{\text{Pitch}}{\text{Total circular divisions}}$$

$$= \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

Observed Reading:

$$\text{Observed} = \text{MSR} + (\text{CSR} \times LC)$$

$$= 3 + (25 \times 0.01) = 3.25 \text{ mm}$$

Applying Zero Correction:

$$\text{True Reading} = \text{Observed Reading} - (\text{Zero Error})$$

$$\text{True} = 3.25 - (-0.05) = 3.30 \text{ mm}$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

6. (3)

According to the laws of physics, the speed of an electromagnetic wave in any medium cannot exceed the speed of light in a vacuum ($c = 3 \times 10^8$ m/s). Let's calculate the speed for each option:

Option (1): $v = \frac{4.5 \times 10^{15}}{1.5 \times 10^7} = 3.0 \times 10^8$ m/s (Possible, traveling in a vacuum)

Option (2): $v = \frac{4.0 \times 10^{14}}{2.0 \times 10^6} = 2.0 \times 10^8$ m/s (Possible, traveling in a denser medium like glass)

Option (3): $v = \frac{5.0 \times 10^{14}}{1.0 \times 10^6} = 5.0 \times 10^8$ m/s (Impossible, as $v > c$)

Option (4): $v = \frac{8.0 \times 10^{15}}{4.0 \times 10^7} = 2.0 \times 10^8$ m/s (Possible, traveling in a denser medium)

Since the speed of the wave in option (3) exceeds 3×10^8 m/s, it violates the fundamental postulates of physics and is therefore not possible.

(NCERT Ed. 2025-26, 12th, Page No. 207)

7. (1)

$$a_{\max} = \omega^2 A = 4\pi^2 \times 2 = 8\pi^2 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 272)

8. (3)

$$|\vec{F}| = i|(\vec{l}_{eff} \times \vec{B})|$$

$$F = (\sqrt{2})\left(2\sqrt{2}\right)(0.1) = 0.4\text{N}$$

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[Contd...

9. (2)

As the point is on the axis of the dipole the magnetic field is along the magnetic dipole moment.

(NCERT Ed. 2025-26, 12th, Page No. 140)

10. (4)

All the statements are correct. In case of vertical circular motion minimum speed at lowest point is $\sqrt{5gl}$ and that at the highest point is $\sqrt{gl}$.

All the constant force are definitely conservative because their work done comes out to be independent of path.

(NCERT Ed. 2025-26, 11th, Page No. 79)

11. (1)

$$x_{COM} = \frac{5\times 2 + 3\times 8}{8} = \frac{17}{4}$$

$$y_{COM} = \frac{5\times 6 + 3\times 2}{8} = \frac{36}{8} = \frac{9}{2}$$

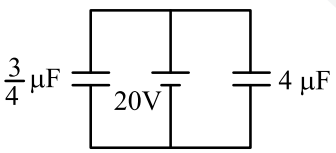
Distance of COM from $\left(\frac{1}{4}, \frac{3}{2}\right)$

$$d = \sqrt{4^2 + (3)^2} = 5\text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

12. (2)

3 μF and 1 μF are in series with equivalent $\frac{3}{4}$ μF



Charge on C_{eq} is $\frac{3}{4}\times 20 = 15\,\mu C$

Charge on 4 μC is 4 × 20 = 80 μC

Total charge = 95 μC

(NCERT Ed. 2025-26, 12th, Page No. 72)

13. (3)

$$\frac{dv}{dt} = 6t$$

$$v - 2 = 3t^2$$

$$v = 3t^2 + 2$$

v = 5 m/s

(NCERT Ed. 2025-26, 11th, Page No. 18)

14. (2)

Stress: Since the wires are connected in series, they are subjected to the same tensile force (F). the stress is the same in both.

Strain: the stress is the same but the Young's Moduli are different ($Y_{brass} \neq Y_{alum}$), the strain will be different in each wire.

(NCERT Ed. 2025-26, 11th, Page No. 171)

15. (2)

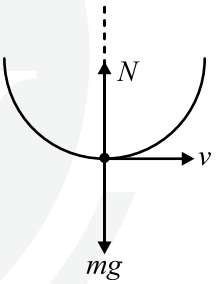
Velocity = slope of position-time graph.

Avg velocity = slope of chord of x-t graph

$v_{avg} > v$

(NCERT Ed. 2025-26, 11th, Page No. 14)

16. (1)



$$N = mg + \frac{mv^2}{R}$$

$$N = 20 + \frac{2\times 400}{5}$$

N = 180 N

(NCERT Ed. 2025-26, 11th, Page No. 79)

17. (1)

$$I_1 = \frac{I_0}{2}$$

$$I_2 = \frac{I_0}{2}\cos^2 45° = \frac{I_0}{4}$$

$$\frac{I_1}{I_2} = 2$$

(NCERT Ed. 2025-26, 12th, Page No. 270)

[Contd...

18. (2)

$$K=\frac{Ar^{2/3}}{B+t^3}\Rightarrow [B]=T^3 \text{ and}$$

$$[A]=[K][B]L^{-2/3}=\left(ML^2T^{-2}\right)\left(T^3\right)L^{-2/3}=[ML^{4/3}T]$$

Let $F=A^xB^yL^z$ have dimensions of force MLT^{-2}.
Matching powers (dimensional analysis):

$$M: x=1; L: \frac{4}{3}x+z=1\Rightarrow z=-\frac{1}{3};$$

$$T: x+3y=-2\Rightarrow y=-1$$

Therefore $x^2-y+z=1-(-1)-\frac{1}{3}=\frac{5}{3}$

(NCERT Ed. 2025-26, 11th, Page No. 8)

19. (1)

$$E_{net}=\frac{Kp}{r^3}\sqrt{1+3\cos^2\theta}$$

$$=\frac{Kp}{125}\sqrt{1+3\left(\frac{16}{25}\right)}$$

$$\Rightarrow E_{net}=\frac{Kp}{625}\times\sqrt{73}$$

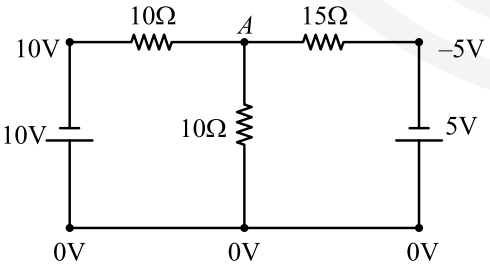
(NCERT Ed. 2025-26, 12th, Page No. 25)

20. (2)

$$v=\sqrt{\frac{T}{\mu}}=\sqrt{\frac{20\times10^{-2}}{2\times10^{-3}}}=10\text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 285)

21. (1)



Let potential at $A = X$
Using nodal analysis at A.

$$\frac{X-10}{10}+\frac{X-0}{10}+\frac{X+5}{15}=0$$

$$3(X-10)+3(X)+2(X+5)=0$$

$$8X-20=0$$

$$X=\frac{20}{8}=\frac{5}{2}=2.5\text{ V}$$

$$\text{Current }=\frac{2.5}{10}=0.25\text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 98)

22. (3)

Magnetic susceptibility is independent of temperature.

(NCERT Ed. 2025-26, 12th, Page No. 148)

23. (2)

For (A): $a = 2$

$\therefore v = 2t + 4 \rightarrow$ Straight line (positive slope)

For (B): $a = t$

$\therefore v=\frac{t^2}{2}+4 \rightarrow$ upward parabola

For (C): $a = -t$

$\therefore v=\frac{-t^2}{2}-4 \rightarrow$ downward parabola

For (D): $a = -2$

$\therefore a = -2t + 4 \rightarrow$ straight line (negative slope)

(NCERT Ed. 2025-26, 11th, Page No. 16)

24. (4)

$$B=\frac{\mu_0 I}{4\pi d}[\sin\theta_1+\sin\theta_2]=\left(\frac{\mu_0 I}{4\pi\times4}[\sin90°+\sin37°]\right)$$

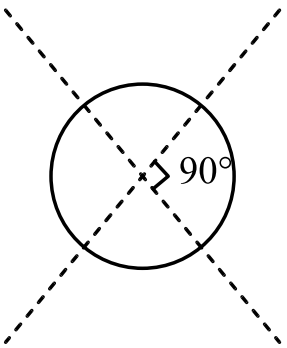
$$=\frac{10^{-7}}{4\times10^{-2}}\times\left[1+\frac{3}{5}\right]=4\times10^{-6}\text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 113)

25. (4)

MOI about z-axis $\frac{MR^2}{2}$

Using perpendicular axis theorem



The moment of inertia about any axis in plane of disc and passing through centre is $\frac{MR^2}{4}$

(NCERT Ed. 2025-26, 11th, Page No. 116)

[Contd...

26. (1)

$M = I(\pi R^2)$

$2\pi R = n(2\pi r)$

$\Rightarrow r = \frac{R}{n}$

$A' = \pi r^2 = \frac{\pi R^2}{n^2}$

$M' = n(I)(\pi r^2) = \frac{n(I)\pi R^2}{n^2} = \frac{M}{n}$

(NCERT Ed. 2025-26, 12th, Page No. 126)

27. (2)

$\phi = BA$

$\frac{d\phi}{dt} = \frac{B\,dA}{dt} + A\frac{dB}{dt} = 0$

The increasing area and decreasing magnetic field may give constant flux.

(NCERT Ed. 2025-26, 12th, Page No. 158)

28. (2)

For an isolated capacitor charge remains constant and insertion of dielectric causes the capacitance to increase therefore the energy stored by the capacitor decreases.

Assertion is false but the reason is true.

(NCERT Ed. 2025-26, 12th, Page No. 74)

29. (2)

$Y = \overline{\overline{AB}} + \overline{B}$

$= AB + \overline{B}$

(NCERT Ed. 2025-26, 12th, Page No. 341)

30. (3)

Let θ be the temperature at 50 cm from hot end.

$\frac{dQ}{dt} = kA\frac{\Delta T}{\Delta x} \Rightarrow \frac{\Delta T}{\Delta x} = \frac{dQ}{dt} \times \frac{1}{kA}$

$\therefore \frac{\Delta T}{\Delta x} = \text{constant}$

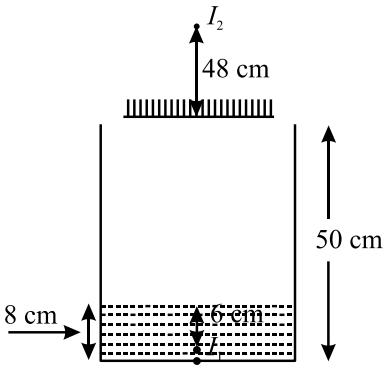
$\therefore \frac{\theta - 100}{0.5} = \frac{20 - 100}{4}$

$\theta - 100 = -10$

$\theta = 90°C$

(NCERT Ed. 2025-26, 11th, Page No. 215)

31. (1)



Apparent depth of

$O = \frac{\text{real depth}}{\text{refractive index}} = \frac{d}{\mu} = 6$

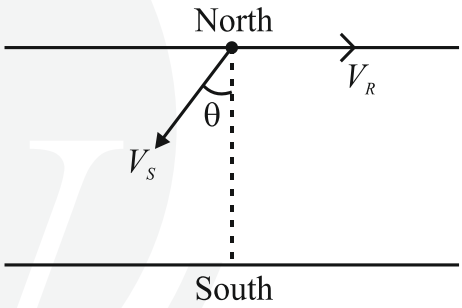
Distance of apparent depth from mirror

= 6 + 42 = 48 cm

Distance between O and I_2 = 48 + 50 = 98 cm

(NCERT Ed. 2025-26, 12th, Page No. 229)

32. (3)



$V_S \sin\theta = V_R$

$20\sin\theta = 10$, θ = 30° (West of South)

(NCERT Ed. 2025-26, 11th, Page No. 34)

33. (4)

Based on the graph:

Wave *A* completes one full cycle in 12 s, so its Time Period $T_A = 12$ s.

Wave *B* completes one full cycle in 8 s, so its Time Period $T_B = 8$ s.

Frequency (f) is inversely proportional to Time Period (T), i.e., $f \propto \frac{1}{T}$.

Therefore, the ratio of their frequencies is:

$\frac{f_B}{f_A} = \frac{T_A}{T_B} = \frac{12}{8} = \frac{3}{2}$.

(NCERT Ed. 2025-26, 11th, Page No. 283)

[Contd...

34. (1)

$$V=\frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r}-\frac{q}{3r}\right]=\frac{q}{6\pi\varepsilon_0 r}$$

So, $E=\frac{1}{4\pi\varepsilon_0}\frac{q}{9r^2}=\frac{6V}{36r}=\frac{V}{6r}$

(NCERT Ed. 2025-26, 12th, Page No. 35)

35. (2)

Induced emf $=n\frac{d\phi_B}{dt}=iR$

$\Rightarrow nd\phi=iRdt\Rightarrow n\Delta\phi=R\Delta Q$

$\Rightarrow \Delta Q=\frac{n\Delta\phi}{R}$

(NCERT Ed. 2025-26, 12th, Page No. 158)

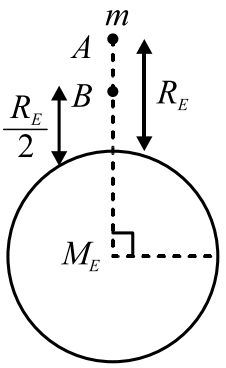
36. (1)

$4t = 0.4 \times 100$

$t = 10$ s

(NCERT Ed. 2025-26, 11th, Page No. 60)

37. (1)



Conservation of mechanical energy

$$-\frac{GM_Em}{2R_E}=-\frac{GM_Em}{\frac{3R_E}{2}}+\frac{1}{2}mv^2$$

$$\Rightarrow \frac{2GM_E}{3R_E}-\frac{GM_E}{2R_E}=\frac{1}{2}v^2$$

$$\Rightarrow \frac{GM_E}{6R_E}=\frac{1}{2}v^2$$

$$\Rightarrow v=\sqrt{\frac{GM_E}{3R_E}}$$

(NCERT Ed. 2025-26, 11th, Page No. 140)

38. (2)

$$\frac{\Delta R_{eq}}{R_{eq}}=R_{eq}\left(\frac{\Delta R_1}{R_1^2}+\frac{\Delta R_2}{R_2^2}\right)$$

$$\Rightarrow \frac{20\times5}{25}\left(\frac{0.1}{25}+\frac{0.4}{400}\right)$$

$$\Rightarrow 4\left(\frac{0.1}{25}+\frac{0.1}{100}\right)$$

$$=4\left(\frac{0.5}{100}\right)=0.02$$

$$\frac{\Delta R_{eq}}{R_{eq}}\times100=2\%$$

(NCERT Ed. 2025-26, 11th, Page No. 6)

39. (2)

At resonance in LCR series a.c. circuit the voltage and current phasor have zero phase difference and the power factor is one.

Statement I is correct but statement II is incorrect.

(NCERT Ed. 2025-26, 12th, Page No. 191)

40. (2)

For the distance r to always increase, the velocity vector ($\vec{v}$) must always have a component pointing away from the origin ($\vec{r}$).

$\vec{r}\cdot\vec{v}\geq0$ for all time t.

Using position $\vec{r}=(ut\cos\theta,ut\sin\theta-\frac{1}{2}gt^2)$ and velocity $\vec{v}=(u\cos\theta,u\sin\theta-gt)$:

$$\vec{r}\cdot\vec{v}=u^2t-\frac{3}{2}ugt^2\sin\theta+\frac{1}{2}g^2t^3\geq0$$

Dividing by t gives a quadratic in t:

$$\frac{1}{2}g^2t^2-\frac{3}{2}ug\sin\theta\cdot t+u^2\geq0$$

$$g^2t^2-3ug\sin\theta t+2u^2\geq0$$

$$g^2\left(t^2-\frac{3u\sin\theta}{g}t\right)+2u^2\geq0$$

Complete the square inside by adding and subtracting $\left(\frac{3u\sin\theta}{2g}\right)^2$:

$$g^2\left[\left(t-\frac{3u\sin\theta}{2g}\right)^2-\frac{9u^2\sin^2\theta}{4g^2}\right]+2u^2\geq0$$

[Contd...

$$g^2\left(t-\frac{3u\sin\theta}{2g}\right)^2-\frac{9}{4}u^2\sin^2\theta+2u^2\geq 0$$

The first term is a squared quantity, so it is always ≥ 0. The lowest possible value for the entire expression occurs when that squared term is exactly zero. For the overall expression to be non-negative at all times, its minimum value must be ≥ 0 :

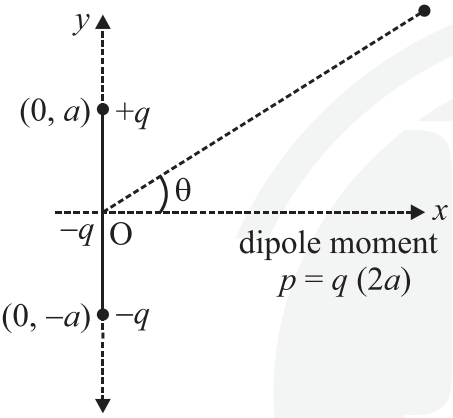
$$2u^2-\frac{9}{4}u^2\sin^2\theta\geq 0$$

$$2\geq\frac{9}{4}\sin^2\theta\Rightarrow\frac{8}{9}\geq\sin^2\theta\Rightarrow\frac{2\sqrt{2}}{3}\geq\sin\theta$$

Hence option (2) correct.

(NCERT Ed. 2025-26, 11th, Page No. 39)

41. (4)



Potential at point P due to charge q (placed at origin) is

$$V_0=\frac{-kq}{r}$$

potential due to remaining charge system (dipole) can be written as

$$V_d=\frac{kp}{r^2}\sin\theta=\frac{kq.2a\sin\theta}{r^2}$$

Hence, $V_p=V_0+V_d$

$$=\frac{kq}{r}\left[\frac{2a\sin\theta}{r}-1\right]$$

(NCERT Ed. 2025-26, 12th, Page No. 51)

42. (4)

Angular velocity, $\omega=\frac{d\theta}{dt}=\frac{d}{dt}\left[\frac{3t^4}{8}-\frac{t^2}{2}\right]$

$$=\frac{3}{8}\times 4t^3-\frac{1}{2}\times 2t=\left(\frac{3}{2}t^3-t\right)\text{rad/s}$$

Angular acceleration, $\alpha=\frac{d\omega}{dt}=\frac{d}{dt}\left[\frac{3}{2}t^3-t\right]$

$$=\frac{3}{2}\times 3t^2-1=\left(\frac{9}{2}t^2-1\right)\text{rad/s}^2$$

Angular acceleration at $2\text{s}=\frac{9}{2}\times 2^2-1=17\text{ rad/s}^2$

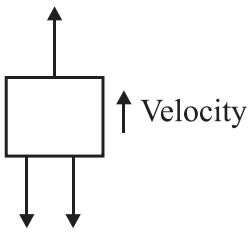
(NCERT Ed. 2025-26, 11th, Page No. 117)

43. (4)

Constant velocity $\Rightarrow a=0$

$\Rightarrow T=W+f$

$= 25000 + 3500 = 28500$ N



$\Rightarrow$ Power = Tv

$= 285000 \times 0.5$

$= 14250$ W

(NCERT Ed. 2025-26, 11th, Page No. 83)

44. (3)

Momentum Conservation: Initial momentum is zero. Therefore, $m_{\text{man}}v_{\text{man}}=m_{\text{woman}}v_{\text{woman}}$

$$80\times 2.5=50\times v_{\text{woman}}$$

$$200=50\times v_{\text{woman}}\Rightarrow v_{\text{woman}}=4.0\text{ m/s}.$$

(Statement 2 is correct, Statement 1 is incorrect).

Work-Energy Theorem:

Work done by the man = Kinetic energy gained by the woman =

$$\frac{1}{2}m_{\text{woman}}v_{\text{woman}}^2=\frac{1}{2}\times 50\times(4.0)^2=25\times 16=400\text{ J}$$

(Statement 3 is correct).

Work done by the woman = Kinetic energy gained by the man

$$=\frac{1}{2}m_{\text{man}}v_{\text{man}}^2=\frac{1}{2}\times 80\times(2.5)^2=40\times 6.25=250\text{ J}$$

Comparing Work: 250 J (done by woman) is less than 400 J (done by man). (Statement 4 is correct).

(NCERT Ed. 2025-26, 11th, Page No. 101)

45. (4)

$$E_x=-\frac{dV}{dx}\Rightarrow E_x=-6x$$

Electric field at $x = 3$ m

$\Rightarrow$ $E_x = -8$ NC^{-1} $\Rightarrow \vec{E}=-8\hat{i}$ NC^{-1}

(NCERT Ed. 2025-26, 12th, Page No. 55)

[Contd...

46. (1)

$W_{irr} = -P_{ext}\Delta V$

$= -2 \text{ bar} \times (0.25 - 0.1) \text{ L} = -2 \times 0.15 \text{ L-bar}$

$= -0.30 \text{ L-bar} = -0.30 \times 100 \text{ J} = -30 \text{ J}$

(NCERT Ed. 2025-26, 11th, Page No. 141)

47. (4)

Order of negative electron gain enthalpy:

S > Se > Te > O

When an electron is added to O, the added electron goes to the smaller n = 2 quantum level and suffers significant repulsion from the other electrons present in this level.

(NCERT Ed. 2025-26, 11th, Page No. 90)

48. (1)

$Ca(OH)_2 \rightleftharpoons Ca^{2+} + 2OH^-$

pH = 9

Hence pOH = 14 – 9 = 5

$[OH^-] = 10^{-5} M$

Hence $[Ca^{2+}] = \frac{10^{-5}}{2}$

Thus $K_{sp} = [Ca^{2+}][OH^-]^2$

$= \left(\frac{10^{-5}}{2}\right)(10^{-5})^2 = 0.5 \times 10^{-15}$

(NCERT Ed. 2025-26, 11th, Page No. 204)

49. (1)

S atom has sp^3d hybridisation

(NCERT Ed. 2025-26, 11th, Page No. 125)

50. (1)

Boiling point is inversely proportional to volatility.

Order of bp = $H_2O > H_2Te > H_2Se > H_2S$

So order of volatility = $H_2O < H_2Te < H_2Se < H_2S$

(NCERT Ed. 2022-23, 12th, Page No. 188)

51. (1)

Lower the value of (i × m), higher will be the freezing point of the solution.

- Glucose does not dissociate, $i = 1$
- $KCl \rightarrow K^+ + Cl^-$, $i = 2$
- $Mg(NO_3)_2 \rightarrow Mg^{2+} + 2NO_3^-$, $i = 3$
- $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$, $i = 5$

(NCERT Ed. 2025-26, 12th, Page No. 24)

52. (4)

- B_{12} vitamin can be stored in our body.
- Lactose is a reducing sugar.

(NCERT Ed. 2025-26, 12th, Page No. 296)

53. (2)

1 mole of $C_6H_{12}O_6$ has $6 \times 6.023 \times 10^{23}$ C atoms

0.35 mole of $C_6H_{12}O_6$ has $0.35 \times 6 \times 6.023 \times 10^{23} = 1.26 \times 10^{24}$ C atoms

(NCERT Ed. 2025-26, 11th, Page No. 18)

54. (1)

$Ca^{2+} + 2e^- \rightarrow Ca$

For 1 mole Ca^{2+}, 2F is required

For 2.5 mole, Ca^{2+}, $\frac{2}{1} \times 2.5 = 5\,F$ is required.

(NCERT Ed. 2025-26, 12th, Page No. 51)

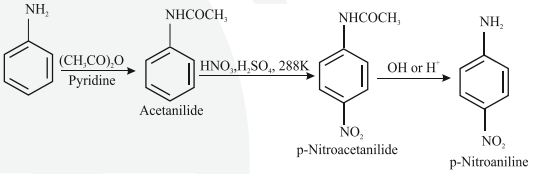
55. (3)

$$\ln 4 = \frac{E_a}{8.314}\left[\left(\frac{1}{300}\right) - \left(\frac{1}{310}\right)\right]$$

$$E_a = \frac{1.386 \times 8.314 \times 300 \times 310}{10} = 107165 \text{ J} = 107.2 \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 81)

56. (1)



(NCERT Ed. 2025-26, 12th, Page No. 273)

57. (1)

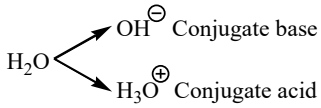
In aqueous solution, electron donating inductive effect, solvation effect (H-bonding) and steric hindrance all together affect basic strength of substituted amines.

Basic character:

$\underset{2^\circ}{(CH_3)_2NH} > \underset{1^\circ}{CH_3NH_2} > \underset{3^\circ}{(CH_3)_3N}$

(NCERT Ed. 2025-26, 12th, Page No. 271)

58. (3)



HF on loss of H^+ ion becomes F^- is the conjugate base of HF

Example: $\underset{\text{Acid}}{HF} + \underset{\text{Base}}{H_2O} \rightleftharpoons \underset{\text{Conjugate base}}{F^\ominus} + \underset{\text{Conjugate acid}}{H_3O^\oplus}$

(NCERT Ed. 2025-26, 11th, Page No. 194)

59. (4)

Size of the atom is bigger, Van der Waal's force is more and boiling point is more

So b.p. of $CH_3CH_2I > CH_3CH_2Br > CH_3CH_2Cl > CH_3CH_2F$

(NCERT Ed. 2025-26, 12th, Page No. 170)

[Contd...

60. (3)

$$\begin{array}{lcccc} & N_2 + 3H_2 & \rightarrow & 2NH_3 \\ & 1 \quad 3 & & 0 \\ \text{At eq} & 0.6 \quad 3-3\times0.4 & & 0.8 \end{array}$$

$0.6 + 1.8 + 0.8 = 3.2$

(NCERT Ed. 2025-26, 11th, Page No. 183)

61. (2)

$(CH_3)_2CHC{\equiv}CH + KMnO_4 \rightarrow (CH_3)_2CHCOOH$

(NCERT Ed. 2025-26, 11th, Page No. 316)

62. (2)

$pH = pK_a + \log\frac{salt}{acid}$; $4.5 = 4.2 + \log\frac{0.2(300-V)}{0.1\,V}$

$1.99\,V = 600 - 2\,V$

$3.99\,V = 600 \Rightarrow V = 150$ mL

(NCERT Ed. 2025-26, 11th, Page No. 203)

63. (3)

$\text{Wave length(electron)} = \frac{h}{mv}$

$\text{Wave length (neutron)} = \frac{h}{1840\,mv^1}$

$v^1 = \frac{h}{1840\,m \times \text{wavelength}} = \frac{v}{1840}$

(NCERT Ed. 2025-26, 11th, Page No. 50)

64. (1)

$CH_3C(Cl_2)CH_3 \xrightarrow{\text{Hydrolysis}} CH_3COCH_3$

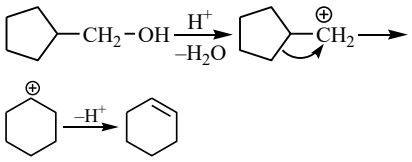
(NCERT Ed. 2025-26, 12th, Page No. 173)

65. (1)

Complexes in which a metal is bound to only one kind of donor groups known as homoleptic.

(NCERT Ed. 2025-26, 12th, Page No. 122)

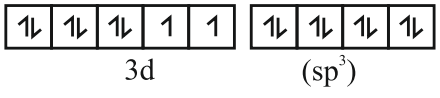
66. (2)



(NCERT Ed. 2025-26, 12th, Page No. 209)

67. (4)

Ni^{2+} in $[Ni(H_2O)_4]^{2+}$:



$\rightarrow 4H_2O$

No. of unpaired $e^- = 2$

$\mu = \sqrt{2(2+2)}$ BM $= \sqrt{8}$ BM

(NCERT Ed. 2025-26, 12th, Page No. 130)

68. (4)

Due to involvement of lone pair of electrons in resonance in phenol, it will have positive charge (partial), hence incoming proton will not be able to attack easily.

(NCERT Ed. 2025-26, 12th, Page No. 210)

69. (2)

$3H_2O \rightarrow 3H^+ + 3OH^-$

$3H^+ + 3e^- \rightarrow \frac{3}{2}H_2$

$3OH^- - 3e^- \rightarrow \frac{3}{4}O_2 + \frac{3}{2}H_2O$

Total gaseous mole $= \frac{3}{4} + \frac{3}{2} = 2.25$

Volume at STP = 2.25 × 22.4 = 50.4 L

(NCERT Ed. 2025-26, 12th, Page No. 53)

70. (2)

A. Iodoform Reaction: Both have CH_3CO- group.
B. Cannizzaro Reaction: Requires no α -hydrogen. Both have α -hydrogens.
C. Aldol Condensation: Requires α -hydrogen. Both have α -hydrogens.
D. Tollen's Test: Positive for aldehydes only.
E. Clemmensen Reduction Reduces C=O to CH_2.

So, both undergo A, C, and E.

(NCERT Ed. 2025-26, 12th, Page No. 242)

71. (2)

- Zn^{2+}: Colourless
- Mn^{2+}: Pink
- Cu^{2+}: Blue
- Fe^{2+}: Green

(NCERT Ed. 2025-26, 12th, Page No. 103)

72. (3)

$C_6H_5-CH_2-NH_2 \xrightarrow{HNO_2} C_6H_5-CH_2-OH$

(NCERT Ed. 2025-26, 12th, Page No. 272)

73. (3)

First order rate constant is given as,

$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]_t}$

99% completed reaction,

$k = \frac{2.303}{t}\log\frac{100}{1} = \frac{2.303}{t}\log 10^2$

$k = \frac{2.303}{t} \times 2\log 10$

$t = \frac{2.303}{k} \times 2 = \frac{4.606}{k}$; $t = \frac{4.606}{k}$

(NCERT Ed. 2025-26, 12th, Page No. 83)

[Contd...

74. (4)

The central silicon atom is sp^3 hybridised, has no lone pair of electrons and hence, the structure should be tetrahedral.

(NCERT Ed. 2022-23, 11th, Page No. 324)

75. (1)

$$\% \text{ of carbon } = \frac{12 \times 0.44 \times 100}{44 \times 0.15} = 80$$

% of hydrogen = (100 – 80) = 20

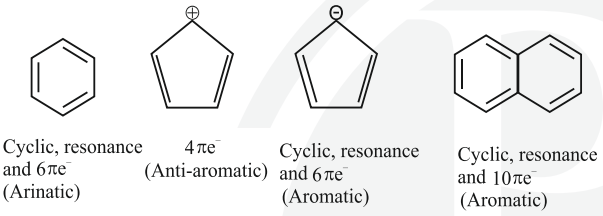
(NCERT Ed. 2025-26, 11th, Page No. 286)

76. (2)

$\Delta H < 0, \Delta S > 0$

(NCERT Ed. 2025-26, 11th, Page No. 162)

77. (2)



(NCERT Ed. 2025-26, 11th, Page No. 321)

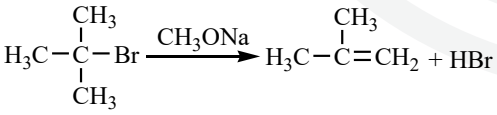
78. (4)

DBE is 1



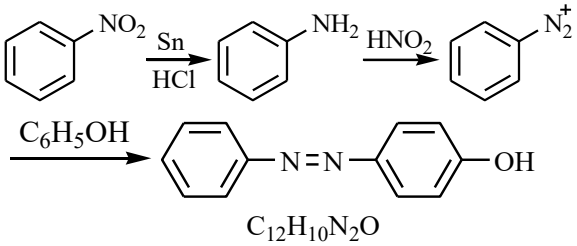
(NCERT Ed. 2025-26, 11th, Page No. 297)

79. (2)



(NCERT Ed. 2025-26, 12th, Page No. 181)

80. (4)



$C_{12}H_{10}N_2O$

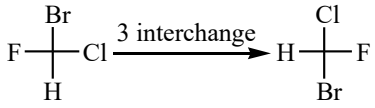
(NCERT Ed. 2025-26, 12th, Page No. 275)

81. (2)

$$HCOOH \xrightarrow[-H_2O]{\text{conc. } H_2SO_4} CO + H_2O$$

(NCERT Ed. 2022-23, 12th, Page No. 200)

82. (1)



So they are enantiomers

(NCERT Ed. 2025-26, 12th, Page No. 177)

83. (2)

It is because of the fact that for spontaneity, the value of $\Delta G = (\Delta H - T\Delta S)$ should be < 0. If ΔS is – ve, the value of $T\Delta S$ shall have to be less than ΔH or the value of ΔS has to be less than $\frac{\Delta H}{T}$ i.e., $\frac{x}{298}$.

(NCERT Ed. 2025-26, 11th, Page No. 161)

84. (2)

CO and C_2^{2-} bond order is 3.

(NCERT Ed. 2022-23, 12th, Page No. 210)

85. (4)

$CH_2=\overset{+}{C}H$ is less stable than $CH_3-\overset{+}{C}H_2$

(NCERT Ed. 2025-26, 11th, Page No. 275)

86. (3)

$$Y_A = \frac{X_A P_A^0}{X_A P_A^0 + X_B P_B^0} = \frac{1}{1 + \frac{X_B P_B^0}{X_A P_A^0}} = \frac{1}{1 + \frac{2}{1} \times \frac{2}{1}} = \frac{1}{5} = 0.2$$

(NCERT Ed. 2025-26, 12th, Page No. 11)

87. (1)

$$(I.E)_H = -(E_1)_H = X$$

$$\Delta E = (E_0 - E_3)_H \times z^2 = 9\left[\frac{-x}{36} - \left(\frac{-x}{9}\right)\right] = \frac{3X}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 48)

88. (1)

Cell constant = Resistance × Specific conductance

$= 200 \times 0.4 \times 10^{-3} = 0.08 \text{ cm}^{-1}$

(NCERT Ed. 2025-26, 12th, Page No. 43)

89. (2)

Metallic conductance decreases with increase of temperature because with increase in temperature vibration of metal atoms increases due to which movement of electrons become difficult.

(NCERT Ed. 2025-26, 12th, Page No. 42)

90. (3)

- Mn^{2+}: Group-IV
- As^{3+}: Group-II B
- Cu^{2+}: Group-II A
- Al^{3+}: Group-III

[Contd...

91. (3)

- The Green Revolution succeeded in tripling the food supply but yet it was not enough to feed the growing human population.
- Increased yields have partly been due to the use of improved crop varieties, but mainly due to the use of better management practices and use of agrochemicals (fertilisers and pesticides).
- However, for farmers in the developing world, agrochemicals are often too expensive, and further increases in yield with existing varieties are not possible using conventional breeding.

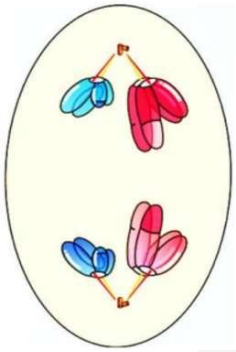
(NCERT Ed. 2025-26, 12th, Page No. 178)

92. (2)

Phloem fibres are made up of sclerenchymatous cells.

(NCERT Ed. 2022-23, 11th, Page No. 88)

93. (3)



Given figure represent the anaphase I stage of the meiosis I.

(NCERT Ed. 2025-26, 11th, Page No. 127)

94. (1)

In the taxonomic hierarchy of animals, classes with a few similar characters are assigned to a higher category called phylum.

(NCERT Ed. 2022-23, 11th, Page No. 7)

95. (1)

- At present, about 30 recombinant therapeutics have been approved for human-use the world over.
- In India, 12 of these are presently being marketed.

(NCERT Ed. 2025-26, 12th, Page No. 181)

96. (4)

- In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B, chains of human insulin and introduced them in plasmids of E. coli to produce insulin chains.
- The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency.
- In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre).
- In 1997, an American company got patent rights on Basmati rice through the US Patent and Trademark Office.

(NCERT Ed. 2025-26, 12th, Page No. 185)

97. (4)

During S phase, if the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 48 chromosomes at G_1, even after S phase the number of chromosomes remains the same, i.e., 48.

(NCERT Ed. 2025-26, 11th, Page No. 121)

98. (4)

Nucleolus is a site for active ribosomal RNA synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 100)

99. (3)

Once the biochemical oxygen demand (BOD) of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called activated sludge.

(NCERT Ed. 2025-26, 12th, Page No. 154)

100. (2)

Equisetum plant belongs to the class sphenopsida.

(NCERT Ed. 2025-26, 11th, Page No. 32)

101. (1)

Inspite of differences in structure and form of different animals, there are fundamental features common to various individuals in relation to the arrangement of cells, body symmetry, nature of coelom, patterns of digestive, circulatory or reproductive systems. These features are used as the basis of animal classification.

(NCERT Ed. 2025-26, 11th, Page No. 37)

[Contd...

102. (1)
Insects have a network of tubes (tracheal tubes) to transport atmospheric air within the body.
(NCERT Ed. 2025-26, 11th, Page No. 183)

103. (3)
O_2 can bind with haemoglobin in a reversible manner to form oxyhaemoglobin. Each haemoglobin molecule can carry a maximum of four molecules of O_2.
(NCERT Ed. 2025-26, 11th, Page No. 189)

104. (2)
Though the seeds differ greatly, the early stages of embryo development (embryogeny) are similar in both monocotyledons and dicotyledons.
(NCERT Ed. 2025-26, 12th, Page No. 19)

105. (2)
The descending limb of Henle's Loop is permeable to water, which allows water to leave the filtrate, concentrating it as it moves down.
(NCERT Ed. 2025-26, 11th, Page No. 209)

106. (3)

Physalia	Portuguese man-of-war
Adamsia	Sea anemone
Pennatula	Sea-pen
Gorgonia	Sea-fan

(NCERT Ed. 2025-26, 11th, Page No. 41)

107. (4)
Ringworm is caused by a fungal disease which is a common infection of the skin and nail. It can cause an itchy red, circular, rash.
(NCERT Ed. 2025-26, 12th, Page No. 133)

108. (4)
Excessive dosage of cocaine causes hallucinations. Other well-known plants with hallucinogenic properties are *Atropa belladona* and Datura.
(NCERT Ed. 2025-26, 12th, Page No. 143)

109. (1)
Na^+/K^+ ATPase pump maintains resting membrane potential by pumping 3 Na^+ out and 2 K^+ in.
Inside axon (axoplasm) K^+ concentration is high Na^+ concentration is low. Endocrine system coordinates body functions via hormones. Thalamus is part of the forebrain (diencephalon), not hindbrain. Emotions like anger and rage are mainly linked to the hypothalamus and limbic system. Electrical synapses are faster due to direct ion flow through gap junctions. Chemical synapses involve neurotransmitter release, so they are slower.
(NCERT Ed. 2025-26, 11th, Page No. 236)

110. (2)
Trichophyton species is an opportunistic fungal disease of mammals. Filariasis is a mosquito borne disease caused by the nematode *wuchereria bancrofti*.
- ***Plasmodian vivax*** is a protozoal parasite and a human pathogen.
- **Pneumonia** is a bacterial disease caused by *streptococcus pneumoniae or Haemophilus influenzae*.

(NCERT Ed. 2025-26, 12th, Page No. 131)

111. (3)
Stone or insoluble mass of crystallised salts (oxalates, etc.) formed within the kidney is symptom of renal calculi.
(NCERT Ed. 2025-26, 11th, Page No. 214)

112. (1)
Aves are homoiothermous (warm-blooded), meaning they can maintain a constant body temperature.
(NCERT Ed. 2025-26, 11th, Page No. 49)

113. (1)
Urine formed by the nephrons is ultimately carried to the urinary bladder where it is stored till a voluntary signal is given by the central nervous system (CNS).
(NCERT Ed. 2025-26, 11th, Page No. 212)

114. (1)
Loose connective tissues has cells and fibres loosely arranged in semi-fluid ground substance.
(NCERT Ed. 2020-21, 11th, Page No. 103)

115. (1)
In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. Viroids was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.
(NCERT Ed. 2025-26, 11th, Page No. 21)

116. (4)
First terrestrial plants to possess vascular tissues - Pteridophytes
(NCERT Ed. 2025-26, 11th, Page No. 30)

117. (4)
Both wind and water pollinated flowers are not very colourful and do not produce nectar.
(NCERT Ed. 2025-26, 12th, Page No. 13)

[Contd...

118. (1)

Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon, and is commonly found in autonomous and the somatic neural systems.

(NCERT Ed. 2025-26, 11th, Page No. 231)

119. (1)

The sex determining mechanism in case of humans is XY type. Out of 23 pairs of chromosomes present, 22 pairs are exactly same in both males and females; these are the autosomes. A pair of X chromosomes are present in the female, whereas the presence of an X and Y chromosome are determinant of the male characteristic.

(NCERT Ed. 2025-26, 12th, Page No. 71)

120. (3)

Diatoms are the chief 'producers' in the oceans.

(NCERT Ed. 2025-26, 11th, Page No. 14)

121. (2)

- **Dura mater:** This is the outermost layer, closest to skull.
- **Arachnoid:** This is the thin middle layer.
- **Pia mater:** Innermost layer of meninges, which is in contact with brain tissue.

(NCERT Ed. 2025-26, 11th, Page No. 235)

122. (1)

A typical nucleosome contains 200 bp of DNA helix.

(NCERT Ed. 2025-26, 12th, Page No. 83)

123. (3)

A. Epidermal tissue system forms the outer-most covering of the whole plant body. This is correct. The primary function of the epidermis is to provide a protective outer layer.

B. It is made up of elongated cells. This is correct. The cells are typically elongated and form a compact, continuous layer.

C. It is always multi-layered. This is incorrect. The epidermis is typically single-layered.

D. Epidermal cells have a large vacuole. This is correct. Epidermal cells are living and contain a large central vacuole, pushing the cytoplasm to the cell periphery.

E. Epidermal cells are parenchymatous. This is correct. The fundamental cell type of the epidermis is parenchymatous, although these cells are highly specialized.

(NCERT Ed. 2025-26, 11th, Page No. 71)

124. (4)

Since DNA fragments are negatively charged molecules they can be separated by forcing them to move towards the anode under an electric field through a medium/matrix. The DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel.

(NCERT Ed. 2025-26, 12th, Page No. 168)

125. (4)

Nicotine stimulates adrenal gland to release adrenaline and nor-adrenaline into blood circulation, both of which raise blood pressure and increase heart rate. Smoking is associated with increased incidence of cancers of lung, urinary bladder and throat, bronchitis, emphysema, coronary heart disease, gastric ulcer, etc. Tobacco chewing is associated with increased risk of cancer of the oral cavity. Smoking increases carbon monoxide (CO) content in blood and reduces the concentration of haembound oxygen. This causes oxygen deficiency in the body.

(NCERT Ed. 2025-26, 12th, Page No. 144)

126. (4)

A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against the pO_2. Oxygen dissociation curve is highly useful in studying the effect of factors like pCO_2, H^+ concentration, etc., on binding of O_2 with Haemoglobin.

(NCERT Ed. 2025-26, 11th, Page No. 189)

127. (3)

Some species of insects and frogs are cryptically-coloured (*camouflaged*) to avoid being detected easily by the predator. Some are poisonous and therefore avoided by the predators.

Predators - Prudent behaviour do not overexploits prey.

(NCERT Ed. 2025-26, 12th, Page No. 198)

128. (4)

UAA, UAG, UGA are stop terminator codons.

(NCERT Ed. 2025-26, 12th, Page No. 96)

129. (4)

The enzyme DNA ligase is used to join the discontinuously synthesized DNA fragments during replication.

(NCERT Ed. 2025-26, 12th, Page No. 90)

[Contd...

130. (4)

In fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. The enzymes, pyruvic acid decarboxylase and alcohol dehydrogenase catalyse these reactions.

Yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent.

(NCERT Ed. 2025-26, 11th, Page No. 157)

131. (2)

Psittacula (parrot) have pneumatic bones and are homoithermous. *Pteropus* (flying fox) is mammal which has mammary gland, is viviparous and homoithermous. So, homeothermy is common feature.

(NCERT Ed. 2025-26, 11th, Page No. 51)

132. (1)

Recombinant DNA	Artificial DNA molecule
Cellulase	Isolation of DNA from plant cell
Ethidium bromide	DNA staining
Agarose	Sea weeds

(NCERT Ed. 2025-26, 12th, Page No. 168)

133. (4)

The repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA.

(NCERT Ed. 2025-26, 12th, Page No. 173)

134. (4)

Operator site	Repressor binding site
Promoter site	RNA polymerase binding site
Structural gene	Encode proteins required for lactose metabolism
Regulator gene	Produces repressor protein

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135. (2)

Cytokinins helps to produce new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation. Cytokinins help overcome the apical dominance. They promote nutrient mobilisation which helps in the delay of leaf senescence. GA_3 is used to speed up the malting process in brewing industry. Ethylene induces flowering in mango.

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136. (1)

Tendons	Attach skeletal muscles to bones
Fibroblasts	Areolar connective tissue
Neuroglial cells	Protect neurons
Ligaments	Attach bones to bones

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137. (2)

Aspergillus niger produces citric acid.

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138. (3)

Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle.

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139. (2)

Main plant body of Liverworts is haploid in the form of a leaf like structure called thalloid plant body.

In place of roots, primitive root like structures are present called as rhizoids.

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140. (3)

Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus).

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141. (1)

Arrangement of sepals or petals in flowers is called the aestivation. When sepals or petals in a whorl just touch one another at the margin, without overlapping, it is said to be valvate as in *Calotropis*.

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142. (4)

Plasmodium is a parasitic protozoan that causes malaria in humans. • *Plasmodium* enters our body in the form of sporozoites by infected *Anopheles* mosquito. Sporozoites are the infective stage of *Plasmodium* parasite that is transmitted to human through bite of an infected female anopheles mosquito. Sporozoites go into liver and multiply and become merozoites. Merozoites are stage of the *Plasmodium* parasite responsible for causing symptoms of malaria. Merozoites invade red blood cell and symptom of malaria like fever, chills, anemia and other.

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[Contd...

143. (4)

Both DNA and RNA are polymers of nucleotides. They both contain nucleotides consisting of a nitrogenous base, a sugar and phosphate group. However, DNA contains deoxyribose sugar and RNA contains ribose sugar.

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144. (1)

Indole-3-acetic acid (IAA) is the most common natural auxin found in plants. It occurs naturally and plays a key role in cell elongation, apical dominance, and root initiation.

Naphthalene acetic acid (NAA) and 2,4-dichlorophenoxyacetic acid (2,4-D) are synthetic auxins widely used in agriculture and horticulture—for example, NAA in preventing premature fruit drop and 2,4-D as a selective herbicide against dicot weeds.

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145. (3)

In Cruciferae, the fruit is a siliqua. In Malvaceae, the fruit can be a capsule or a berry.

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146. (3)

The activated toxin binds to the surface of midgut epithelial cells and create pores that cause cell swelling and lysis and eventually cause death of the insect.

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147. (4)

Dysentery, plague, diphtheria, etc., are some of the other bacterial diseases in man.

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148. (1)

X (Patent): A patent is a right granted by a government to an inventor to prevent others from using, making, or selling their invention without permission.

Y (Biopatent): When patents are granted specifically for biological entities (like genes, microorganisms, plants, or animals) and for products derived from them, they are called biopatents.

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149. (3)

When we analyse the species area relationship at large area like the entire continents we will find that the slope of the line to be much stepper and the value of Z would be 0.6 to 1.2.

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150. (3)

Ciliated epithelium is found in the inner linings of hollow organs such as fallopian tubes and bronchioles. It's function is to move particles or mucus in a specific direction over it and the shape of cells in ciliated epithelium can be cuboidal or columnar.

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151. (3)

- They may or may not possess cell wall.
- They do not have membranous inclusion bodies.

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152. (2)

A recombinant DNA is inserted within the coding sequence of an enzyme, β-galactosidase. This results into inactivation of the gene for synthesis of this enzyme, which is referred to as insertional inactivation. The presence of a chromogenic substrate gives blue coloured colonies if the plasmid in the bacteria does not have an insert. Presence of insert results into insertional inactivation of the β-galactosidase gene and the colonies do not produce any colour, these are identified as recombinant colonies.

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153. (3)

Haemophilia will occur in 50% daughters when when father is haemophilic and mother is carrier.

Parent- $X^hX^h \times X^hY$

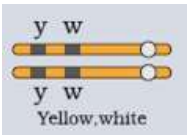
Daughter- X^hX^h (All daughters will be affected)

Parent- $X^hX \times X^hY$

Daughter- X^hX^h (50% daughters will be affected)

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154. (3)



represents female with recessive linked genes.

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155. (1)

- Lion and panther belong to same phylum Chordata.
- Species is the lowest category.
- Dog and cat belong to the different families.

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[Contd...

156. (4)

When ready-made antibodies are directly given to protect the body against foreign agents, it is called passive immunity.

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157. (2)

Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking. While wheezing can be a symptom of various respiratory conditions, it is more commonly associated with asthma and bronchitis rather than emphysema specifically.

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158. (3)

Introducing the functional ADA gene at an early embryonic stage allows it to be incorporated into all cells of the developing individual, potentially providing a permanent cure.

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159. (3)

The intercellular material of cartilage is solid and pliable and resists compression.

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160. (3)

Atrial Natriuretic factor	Vasodilation
Renin	Juxtaglomerular cells
Angiotensin II	Vasoconstrictor
Aldosterone	Adrenal cortex

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161. (4)

The World Summit on Sustainable Development was held in Johannesburg in 2002, where 190 countries pledged their commitment to reducing biodiversity loss by 2010.

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162. (3)

HIV/AIDS is a retrovirus. It uses reverse transcriptase to form DNA from RNA. No complete cure exists for HIV/AIDS. Antiretroviral therapy (ART) controls but does not eliminate the virus. HIV/AIDS is not airborne.

It spreads through:

- Unprotected sexual contact
- Contaminated blood
- Mother to child transmission

WHO focuses on awareness, prevention, testing, and safe practices.

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163. (4)

- Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as *lepidopterans* (tobacco budworm, armyworm), *coleopterans* (beetles) and *dipterans* (flies, mosquitoes).
- B. *thuringiensis* forms protein crystals during a particular phase of their growth.
- These crystals contain a toxic insecticidal protein.

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164. (2)

In a typical anatropous ovule, the body of the ovule fuses with funicle in the region called hilum.

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165. (4)

Cytokinesis is not similar in plant and animal cells. In animals the cell division takes place via formation of cell furrow.

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166. (2)

The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. The thymus is a lobed organ located near the heart and beneath the breastbone.

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167. (4)

Residual Volume (RV): Volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL.

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168. (1)

- There is a linear relationship between incident light and CO_2 fixation rates at low light intensities.
- Light saturation occurs in photoynthesis at 10 per cent of the full sunlight.

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[Contd...

169. (4)

The side-effects of the use of anabolic steroids in female include masculinisation (features like males), increased aggressiveness, mood swings, depression, abnormal menstrual cycles, excessive hair growth on the face and body, enlargement of clitoris, deepening of voice. In males it includes acne, increased aggressiveness, mood swings, depression, reduction of size of the testicles, decreased sperm production.

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170. (2)

Platyhelminthes have an incomplete digestive system, characterized by a single opening to the outside that functions as both the mouth and the anus.

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171. (4)

For glycolysis to continue, cells will require NAD^+ and hexokinase.

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172. (1)

- During urine formation, the tubular cells secrete substances like H^+, K^+ and ammonia into the filtrate.
- Tubular secretion is not the first step in urine formation but glomerular filtration is the first step in urine formation.

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173. (4)

Metacentric chromosome	Centromere in the middle forming two equal arms of chromosomes
Acrocentric chromosome	Centromere situated close to the end forming one extremely short and one very long arms
Sub-metacentric chromosome	Centromere slightly away from the middle forming one shorter arm and one longer arm
Telocentric chromosome	Centromere at the terminal end

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174. (1)

While the membrane or lamellae of the grana have both PS I and PS II the stroma lamellae membranes lack PS II as well as NADP reductase enzyme.

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175. (4)

The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter. Of this, despite occupying about 70 per cent of the surface, the productivity of the oceans are only 55 billion tons. Rest of course, is on land 115 billion tons.

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176. (2)

Dorsiventral leaves are present in dicots.

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177. (4)

Citrate synthase is the enzyme responsible for catalysing the first step of the citric acid cycle (Krebs cycle). It facilitates the condensation of oxaloacetate and acetyl-CoA to form citric acid (citrate).

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178. (1)

The ovary is said to be inferior as in flowers of guava and cucumber, and the ray florets of sunflower.

The ovary is said to be superior as in flowers of mustard,brinjal and china rose.

The ovary is said to be half inferior as in flowers of plum, peach and rose.

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179. (1)

The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as mineralisation.

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180. (1)

The medulla contains centres which control respiration, cardiovascular reflexes and gastric secretions. The hypothalamus is the basal part of diencephalon (forebrain) which contains several groups of neurosecretory cells called nuclei that produce hormones. These hormones regulate the synthesis and secretion of pituitary hormones.

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